# Construction and Validation of a Scale to Measure Celebrity Endorsers' Perceived Expertise, Trustworthiness, and Attractiveness

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*The purpose of this study was to develop a scale for measuring celebrity endorsers' perceived expertise, trustworthiness, and attractiveness. Accepted psychometric scale-development procedures were followed which rigorously tested a large pool of items for their reliability and validity. Using two exploratory and two confirmatory samples, the current research developed a 15-item semantic differential scale to measure perceived expertise, trustworthiness, and attractiveness. The scale was validated using respondents' self-reported measures of intention to purchase and perception of quality for the products being tested. The resulting scale demonstrated high reliability and validity.*

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Marketing and advertising practitioners share the belief that a communicator's character has a significant effect on the persuasiveness of the message. In testimonial advertising, consumers traditionally have been chosen as product endorsers because of their similarity to target audiences. Although this practice continues, a more noticeable trend appears to be endorsements by actors/actresses, athletes, and other celebrities and well-known athletes, who are closely associated with both the product and the target audience (*Business Week* 1987; Miller 1989; Morrison 1980; Slinker 1984).

The selection of an appropriate spokesperson for a product or a service is an important, yet difficult, decision. Is an effective and credible spokesperson someone who is attractive, trustworthy, or an expert, or even a combination of all three traits? Is a credible spokesperson an individual who is dynamic, qualified, authoritative, sociable, or safe? Since Aristotle's time (or before), politicians, orators, and public speakers have attempted to identify the determinant qualities of effective speakers (Giffin 1967).

A number of empirical investigations have examined the effectiveness of using credible spokespersons to enhance the persuasiveness of messages. Studies have measured the process by which a communicator's perceived attractiveness, trustworthiness, and expertise mediate immediate and delayed attitude change and persuasion (Anderson and Clevenger 1963; Baker and Churchill, Jr. 1977; Hovland and Weiss 1951; Johnson, Torcivia, and Poprick 1968; Kelman and Hovland 1953; Patzer 1983; Simon, Berkowitz, and Moyer 1970; Whittaker and Meade 1968).

Several researchers in the field of speech communication have utilized factor analytic techniques to uncover the perceptual structure of source credibility (Applbaum and Anatol 1972; Berlo, Lemert, and Mertz 1969; Bowers and Phillips 1967; McCroskey 1966; Whitehead 1968). Their attempts have resulted in the development of scales, each of which includes a different set of dimensions for the measurement of source credibility. For example, Berlo, Lemert, and Mertz (1969) define source credibility as encompassing the dimensions of safety, qualification, and dynamism. On the other hand, McCroskey (1966) identifies authoritativeness and character as other dimensions of source credibility; while

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Whitehead (1968) identifies objectivity as another dimension of source credibility. In the process of developing scales to measure the effectiveness of celebrity endorsers, a number of researchers in advertising and marketing have expanded the number of dimensions encompassing the source-credibility construct (DeSarbo and Harshman 1985; Simpson and Kahler 1980-81; Wilding and Bauer 1968; Wynn 1987). Table 1 presents a summary of major research studies that have addressed the scaling of source credibility.

Although all the studies were designed to measure the same construct, there is no consistency among the authors as to the number and types of dimensions that source credibility comprises. Furthermore, with the exception of McCroskey (1966), none of the authors have assessed the reliability and validity of the resulting scales. As should be apparent, most attempts to assess the impact of source credibility have been based on instruments of unknown reliability. This fact partially explains the inconsistencies in the literature regarding the impact of communicator credibility as it relates to attitude formation and attitude change. Given the accumulative nature of research, and the fact that researchers

TABLE 1
Source Credibility Scales

| *Author(s)* | *Dimensions Measured* | *Number of Items* | *Reliability Checks* | *Validity Checks* | *Scale* Type* | *Method of Analysis* |
|---|---|---|---|---|---|---|
| Applbaum and Anatol (1972) | Trustworthiness | 13 | No | No | SD | --- |
| | Expertness | 10 | | | | |
| | Dynamism | 5 | | | | |
| | Objectivity | 3 | | | | |
| Berlo, Lemert, and Mertz (1969) | Safety | 5 | No | No | SD | Factor Analysis |
| | Qualification | 5 | | | | |
| | Dynamism | 5 | | | | |
| Bowers and Phillips (1967) | Trustworthiness | 7 | | | SD | Factor Analysis |
| | Competence | 5 | | | | |
| DeSarbo and Harshman (1985) | Expertness | 4 | No | No | SD | --- |
| | Attractiveness | 2 | | | | |
| | Trustworthiness | 1 | | | | |
| | Likability | 2 | | | | |
| | **Additional Dimensions** | | | | | |
| | Evaluative | | | | | |
| | Potency | | | | | |
| | Activity | | | | | |
| McCroskey (1966) | Authoritativeness | 6 | Yes | Yes | SD | Factor Analysis |
| | Character | 6 | | | | |
| | Authoritativeness | 23 | Yes | Yes | LIK | Factor Analysis |
| | Character | 20 | | | | |
| Simpson and Kahler (1980-81) | Believability | 8 | No | Limited | SD | Factor Analysis |
| | Dynamism | 6 | | | | |
| | Expertness | 7 | | | | |
| | Sociability | 3 | | | | |
| Whitehead (1968) | Trustworthiness | 18 | No | No | SD | Factor Analysis |
| | Competence | 4 | | | | |
| | Dynamism | 3 | | | | |
| | Objectivity | 3 | | | | |
| Wynn (1987) | Expertness | 12 | No | No | SD | Factor Analysis |
| | Dynamism | 6 | | | | |
| | Believability | 3 | | | | |
| | Sociability | 3 | | | | |

*SD = Semantic Differential Scale, LIK = Likert Scale

base and build the findings of their studies on that of others, there must be a consistent measurement approach for source credibility. This measurement approach first must provide a theoretical basis for the selection of constructs to represent the hypothesized dimensions of source credibility, and second, must produce a valid, reliable measurement scale. In view of the widespread theoretical and empirical interest in the concept of source credibility, the purpose of the present research is to advance and then to assess a tri-component construct using psychometrically accepted procedures to produce a reliable and valid scale.

## Definitions of Source Credibility

"Source credibility" is a term commonly used to imply a communicator's positive characteristics that affect the receiver's acceptance of a message. Understanding and defining source credibility in the advertising and speech communication context is often confusing because of the many different operationalizations that appear in the literature. For example, in experimental studies, source credibility is often considered a categorical variable, such that individuals are presented as having high or low credibility (e.g., Anderson and Clevenger 1963; Griffitt 1966; Maddux and Rogers 1980). Other approaches commonly used to describe this phenomenon include the use of such labels as: ethos, prestige, reputation, status, authority, competence, etc. (e.g., Applbaum and Anatol 1972; Giffin 1967; McCroskey 1966).

Research and reflection on the topic of celebrity endorsement rest on two general models: the source-credibility model and the source-attractiveness model. The source-credibility model resulted from a landmark study by Hovland and his associates (1953). They analyzed the factors leading to the perceived credibility of the communicator and concluded that two factors—namely, *expertness* and *trustworthiness*—underscore the concept of source credibility. Hovland, Janis, and Kelley (1953) defined expertise as "the extent to which a communicator is perceived to be a source of valid assertions," and trustworthiness as "the degree of confidence in the communicator's intent to communicate the assertions he considers most valid."

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The source-attractiveness model has its origins in the social psychological research and is a component of the "source valence" model of McGuire (McGuire 1985). The attractiveness model contends that the effectiveness of a message depends on source's "familiarity," "likability," "similarity," and "attractiveness" to the respondent.

For the present research, both the source-credibility model of Hovland, et al. (1953), and the attractiveness model of McGuire (1958) were used in defining the dimensions of source valence. Expertise and trustworthness as suggested by Hovland, et al. (1953) (also as the credibility dimension of the McGuire source-valence model), and attractiveness were used as hypothesized dimensions of source attributes. The decision to use attractiveness was further motivated by the fact that attractiveness has become an important factor through the increasing use of celebrities as endorsers for products, services and/or social causes (Baker and Churchill 1977; Caballero, Lumpkin, and Madden 1989; Caballero and Solomon 1984; DeSarbo and Harshman 1985; Patzer 1983). Source likability and similarity were not used in the development of the scale but were employed as measures for nomological validity. Following is a discussion of the three dimensions of expertise, trustworthiness, and physical attractiveness as the hypothesized dimensions of celebrity endorsers' credibility.

**Trustworthiness.** The trust paradigm in communication is the listener's degree of confidence in, and level of acceptance of, the speaker and the message. Giffin (1967) reviewed the concept of trust, in a tour of the centuries from Aristotle to King, and concluded that what Aristotle called "ethos," and what Hovland, Janis, and Kelley (1953) called "source credibility," are the same concept: a listener's trust in a speaker. Furthermore, such terms as "favorable disposition," "acceptance," "psychological safety," and "perceived supportive climate" are often mentioned as favorable consequences of trust (Giffin 1967).

Numerous studies support the effect of trustworthiness on attitude change. For example, in the context of fear-arousing communications, Miller and Baseheart (1969) investigated the impact of source trustworthiness on the persuasibility of the communication. The results indicated that when the communicator was perceived to be highly trustworthy, an opinionated message was more effective than a non-opinionated communication in producing attitude change. However, when trustworthiness was low, this relationship was not significant. McGinnies and Ward (1980) manipulated a source's expertise and trustworthiness to assess the impact of each of these components on the communicator's persuasiveness. Their findings indicated that a source who was perceived to be both an expert and trustworthy generated the most opinion change. In fact, the trustworthy communicator was persuasive, whether an expert or not. Further, Friedman and Friedman (1976), and Friedman, Santeramo, and Traina (1979) investigated several correlates of trustworthiness and concluded that celebrities who are liked will also be trusted. In addition, celebrity

trustworthiness was highly correlated with a respondent's perceived similarity to the source, the level of source's expertise, and the source's attractiveness.

In summary, trustworthiness of the communicator (celebrity) is an important construct in persuasion and attitude-change research. Therefore, a reliable measurement of this construct requires a series of items, rather than the typical single item commonly used to measure the variable as a trustworthy-untrustworthy dichotomy.

**Expertise.** Expertise is the second dimension of source credibility as defined by Hovland, Janis, and Kelley (1953). This dimension is also referred to as "authoritativeness" (McCroskey 1966), "competence" (Whitehead 1968), "expertness" (Applbaum and Anatol 1972), or "qualification" (Berlo, Lemert, and Mertz 1969). Adjectives such as "trained-untrained," "informed-uninformed," and "educated-uneducated" commonly have been used to measure this dimension.

Research investigating source expertise in persuasive communication generally indicates that the source's perceived expertise has a positive impact on attitude change (Horai, Naccari, and Fatoullah 1974; Maddux and Rogers 1980; Mills and Harvey 1972; Ross 1973). For example, Crisci and Kassinove (1973) investigated the effect of the perceived level of communicator expertise ("Dr." versus "Mr.") and the strength of advice (positive versus neutral) on behavioral compliance. The results of this study indicated that respondents' compliance with the source's recommendations directly varied with the perceived level of expertise and the strength of advice. Similarly, Crano (1970) experimentally manipulated the dimensions of expertise and found that subjects exposed to an expert source exhibited more agreement with the advocated position than did those exposed to a low-expertise source. Finally, in a selling context, an expert salesperson induced a significantly higher number of customers to purchase a product than did the nonexpert salesperson (Woodside and Davenport, Jr. 1974).

**Attractiveness.** A considerable body of research in advertising and communication suggests that physical attractiveness is an important cue in an individual's initial judgment of another person (Baker and Churchill 1977; Chaiken 1979; Joseph 1982; Kahle and Homer 1985; Mills and Aronson 1965; Widgery and Ruch 1981). Despite the vast quantity of literature addressing physical attractiveness, the issue is far from clear. A review of the area indicates that the construct of attractiveness is not uni-dimensional and that there are myriad definitions used to operationalize attractiveness. For example, the construct has been defined both in terms of facial and physical attractiveness (Baker and Churchill 1977; Caballero and Solomon 1984; Patzer 1983), with physical attractiveness operationalized in terms of model attractiveness (attractive-unattractive) (Baker and Churchill 1977; Kahle and Homer 1985), chicness (Mills and Aronson 1965), sexiness (Steadman 1969), or sexualness and likability (Maddux and Rogers 1980).

In an exhaustive review, Joseph (1982) summarized the experimental evidence in advertising and related disciplines regarding physically attractive communicators' impact on opinion change, product evaluation, and other dependent measures. He concluded that attractive (versus unattractive) communicators are consistently liked more and have a positive impact on products with which they are associated. Except for a few studies (Mills and Aronson 1965; Maddux and Rogers 1980), Joseph's findings are consistent with others that report that increasing the communicator's attractiveness enhances positive attitude change (Simon, Berkowitz, and Moyer 1970; Kahle and Homer 1985).

## Summary

The review of source-credibility literature provides evidence that credible sources are more persuasive than are sources of low credibility. Additionally, research has shown that highly credible sources induce more behavioral compliance than do less-credible sources (Ross 1973; Woodside and Davenport, Jr. 1974, 1976). However, it is important to recognize that highly credible sources are not always more effective than less-credible ones. In particular, when the audience is already favorably predisposed to the message, a less-credible source can induce greater persuasion than can a highly credible source (Sternthal, Dholakia, and Leavitt 1978). Furthermore, research dealing with the interaction of source and audience characteristics (such as level of authoritarianism or issue involvement) do not always report the greater effectivenesss of higher-credibility sources (Johnson and Izzett 1972; Johnson, Torcivia, and Poprick 1968).

**. . . it is important to recognize that highly credible sources are not always more effective than less-credible ones.**

With the increased use of celebrities in advertising, a valid instrument measuring a celebrity endorser's credibility is essential for understanding the impact of using such individuals in advertising. As in other forms of persuasive communication, advertisers' primary goals are to persuade their audience and to induce an attitude change toward their offerings (Walley 1987).

The following discussion presents the steps for the development of a tri-component celebrity endorser's credibility scale. It includes item generation and reduction, exploratory and confirmatory studies, and reliability and validity of the final subscales.

## Research Methodology

**Development of Items for the Source-Credibility Scale.** In the initial phase

of this research, the literature in the areas of psychology, mass communication, and advertising was reviewed to identify words, phrases, or adjectives used in measuring the traits associated with credible sources. In addition, a large pool of adjectives describing personality traits was developed by consulting previously available sources (Allport and Odbert 1936; Anderson 1968; Osgood, Suci, and Tannenbaum 1957). An effort was made to extract all entries that were likely to be useful in the development of a celebrity endorser's credibility scale. This process resulted in 182 adjectives or descriptor words. The 182 adjectives were screened and reduced to a list of 139, using the following criteria: Extreme words, such as "ferocious" and "boastful," and words denoting temporary states, such as "aghast" and "hurt," were eliminated, since they were not suitable for the impression-formation task.

It was believed that some of the 139 words retained could be unfamiliar to survey respondents. To cull the unfamiliar adjectives or descriptor words, the list was rated by 38 college students. Based on their knowledge of the meaning of each word, the students were instructed to rate each word as either familiar or unfamiliar. Words that were rated unfamiliar by more than 25 percent of the subjects were eliminated. This process reduced the list to a smaller set of 104 words.

In the final editing phase, 52 college students were presented with definitions of trust, expertise, and attractiveness. They were then instructed to carefully study each adjective or descriptor word and to indicate if the item belonged to any of the three dimensions defined earlier. Items with 75 percent or more agreement as belonging to a certain construct were thus retained for further analysis. This procedure reduced the list to 72 words.

**Celebrity/Source Selection Procedure.** To identify an appropriate list of personalities, 40 college students were instructed to list all the celebrity names they could remember in three minutes. The celebrity names were later ranked based on the frequency of mention, categorized by gender, and classified as to whether they had ever participated in a paid commercial. The most frequently mentioned names for each gender, among those who had previously endorsed a product, were John McEnroe and Linda Evans. Celebrities most often mentioned, who had not been involved in advertisements, were Tom Selleck and Madonna.

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In the next phase, 38 college students were asked to indicate their level of familiarity with each of the four celebrities mentioned above. In addition, they were asked to specify the most- and the least-appropriate products that these individuals could sponsor. All the students in the sample showed familiarity with each celebrity. Furthermore, the results suggested that Linda Evans would be most suitable for promoting a line of female cosmetics or a perfume and least appropriate for promoting items such as cigarettes, alcohol, or "fatty foods." Tom Selleck was perceived as appropriate for promoting sports cars and men's cologne and as incompatible with promoting credit cards or long-distance telephone services. The sample indicated tennis rackets and other sporting equipment as items that Jonn McEnroe should promote. Cigarettes, clothing, and drug products were not considered appropriate for him. Finally, for Madonna, the sample indicated that she should promote such items as designer jeans and other modern clothing, but should not promote cars or breakfast cereals.

Using the above results as a guide, it was decided to include frequently purchased products used by a wide cross-section of the population. Thus, the final product/celebrity list included Linda Evans promoting a new perfume; Madonna, a new line of designer jeans; John McEnroe, a line of tennis rackets; and Tom Selleck, a new line of men's cologne.

The next phase of the study was to purify and validate the celebrity endorser's credibility scale. This task was accomplished in two stages. In the exploratory phase, a convenient student sample was employed to further reduce the number of items and to refine the structure of the scale. In the confirmatory phase, two adult samples were used to finalize the list of items in the scale and to establish its reliability and validity.

**Study 1—The Exploratory Phase.** In the exploratory phase of this research, questionnaires for Madonna and John McEnroe were developed, each of which contained two parts. The first section asked respondents to indicate whether or not they were familiar with a particular celebrity and could identify the person as being associated with a show or a specific profession. Additionally, the respondents were asked to supply some standard demographic information. At this stage, if a respondent were to fail to recognize the celebrity, s/he would be removed from the study. In the next section, the respondents were presented with a scenario in which a celebrity endorsed a product; they were then asked to evaluate the celebrity, given 72 semantic differential items discussed in the *Item Development* section of this paper. Two hundred fifty students in a southern university completed the first version of the questionnaire, in which Madonna was the celebrity promoting a new brand of designer jeans. In the second version of the instrument, respondents evaluated John McEnroe promoting a line of tennis rackets. A different group of 240 students completed the second version. Except for changing the name of the celebrity, the product endorsed,

and the gender adjectives (such as "beautiful" to "handsome"), the questionnaires remained identical.

**Exploratory Factor Analysis.** To assess the structure of the source-credibility scale, all the items in the questionnaire were factor analyzed, using the principal components analysis followed by a varimax rotation. The initial factor solution for the sample evaluating Madonna resulted in four factors with eigenvalues greater than one. The four-factor solution accounted for 68.4 percent of the variance. In order to purify the list, items with loadings of 0.3 or greater on more than one of the factors were eliminated. The reduced list was factor analyzed a second time. This resulted in three factors with eigenvalues greater than one, while accounting for 61 percent of the variance. The same steps were repeated for the sample evaluating John McEnroe. Table 2 presents the results of the final factor analysis for both the Madonna and McEnroe data.

As can be seen from Table 2, the first 11 adjectives of the first factor identify the expertise dimension. Adjectives such as "expert," "knowledgeable," "experienced," and "qualified"—all of which have been found to be clear indications of expertise (Applbaum and Anatol 1972; Simpson and Kahler 1980-81; Wynn 1987)—have loadings of 0.6 or higher on that factor. The second factor, consisting of eight items, measures the trustworthiness dimension. Again, items such as "trustworthy," "honest," "dependable," "reliable," and "ethical," which have previously been used to represent this factor, have high loadings (Bowers and Phillips 1967; Whitehead 1968). The third dimension consists of eight items describing attractiveness. Adjectives such as "attractive," "beautiful," "charismatic," and "sophisticated" are used to operationalize this dimension (Baker and Churchill 1977; DeSarbo and Harshman 1985; Patzer 1983). These patterns appear fairly consistent for both the Madonna and McEnroe samples, giving further credence to the stability of these factors.

The final items derived from each factor analysis were tested for their reliability by submitting them to item analysis using item-to-total correlations. The items for each subscale were analyzed separately. To obtain a practical size scale (five items per factor), items with the lowest item-to-total correlations were deleted while maintaining an acceptable level of reliability as

TABLE 2
Factor Loadings for the Three Dimensions of Source Credibility

| | *Factor Loadings* | | | | | |
|---|---|---|---|---|---|---|
| | *Madonna Sample (n = 249)* | | | *John McEnroe Sample (n = 237)* | | |
| | *1* | *2* | *3* | *1* | *2* | *3* |
| Authoritative | .617 | −.037 | .051 | .654 | −.156 | .009 |
| Compatible with the product | .795 | .103 | .070 | .706 | −.147 | .022 |
| Expert | .765 | .107 | .045 | .701 | −.037 | .056 |
| Informative | .812 | .170 | .183 | .747 | .108 | .027 |
| Experienced | .725 | .186 | .021 | .531 | .077 | .034 |
| Intelligent | .697 | .262 | .119 | .699 | .133 | .054 |
| Informed about the product | .799 | .212 | .055 | .631 | −.025 | −.195 |
| Knowledgeable | .798 | .246 | .174 | .695 | .121 | .079 |
| Qualified | .869 | .069 | .131 | .536 | .116 | −.141 |
| Familiar with the product | .696 | .071 | .132 | .606 | −.020 | −.233 |
| Skilled | .699 | .042 | .186 | .607 | −.061 | .054 |
| Dependable | .159 | .742 | .150 | .042 | .683 | −.013 |
| Fair | −.005 | .655 | .198 | .108 | .709 | .188 |
| Reliable | .234 | .780 | .185 | .109 | .739 | .006 |
| Sincere | .152 | .720 | .206 | .002 | .609 | .223 |
| Trustworthy | .128 | .768 | .215 | .169 | .772 | .158 |
| Truthful | .211 | .704 | .185 | .108 | .744 | .146 |
| Honest | .139 | .664 | .280 | .142 | .639 | .251 |
| Ethical | .241 | .587 | .227 | .178 | .646 | .153 |
| Attractive | .046 | .234 | .831 | −.051 | .158 | .721 |
| Classy | .206 | .162 | .646 | .049 | .196 | .500 |
| Sophisticated | .189 | .275 | .596 | .052 | .175 | .496 |
| Handsome/Beautiful | .154 | .252 | .773 | .054 | .090 | .692 |
| Glamourous | .149 | .202 | .761 | .079 | .187 | .541 |
| Elegant | .212 | .185 | .661 | −.184 | .059 | .704 |
| Sexy | .107 | .248 | .744 | −.065 | .093 | .761 |
| Charming | .170 | .283 | .691 | −.112 | .261 | .597 |

measured by Cronbach's alpha (Peter 1979). To determine if the subscales were equally reliable for different celebrities and genders, Cronbach's alpha was computed for both male and female respondents for Madonna and John McEnroe. The results indicated a highly reliable scale. Both male and female respondents had equally reliable response patterns, and the total sample for each subscale and celebrity had a reliability coefficient of 0.8 or higher.

**Study 2—The Confirmatory Analysis.** In the confirmatory phase of the study, the final scale's reliability and validity were assessed by using the 15 items (five items per subscale) obtained from the exploratory phase, along with several other validation items. These additional items were included to measure respondents' likelihood to *inquire about, consider purchasing,* and *actually purchase* for personal use the product sponsored by the particular celebrity. Inquiry, consideration, and purchase represent increasing levels of commitment toward the product. These criterion variables are commonly used to measure the effectiveness of source credibility in marketing research (Baker and Churchill 1977; Kahle and Homer 1985; Rubin, Mager, and Friedman 1982). For further validation of the scale, respondents were asked to indicate the likelihood that they would *inquire about, consider purchasing,* or *actually purchase* the product as a gift. Because of a gift's conspicuous nature, gift-giving decisions, as compared to purchase for self, are often perceived as being more important and more involving (Belk 1982; Clarke and Belk 1979; Kassarjian 1981).

Additionally, questions were included to assess respondents' perceived similarity to and likability of the celebrity. Similarity and likability have been used extensively in the literature as determinants of identification and interpersonal attraction between the source and the message recipient (Aronson and Worchel 1966; Berscheid 1966; Griffitt 1966; Kelman 1961). Finally, respondents were asked to evaluate the celebrity's compatibility with the product, the celebrity's role in helping both the image and the sales of the product, and the perceived comparative distinctiveness of that product as compared to other brands in the market.

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The 15 adjectives developed in the exploratory phase of this research, along with the validation behaviors discussed above, were administered to an adult sample. Furthermore, in the confirmatory phase, different celebrities, specifically, Linda Evans and Tom Selleck, were used to support the generalizability of the scale. Linda Evans was endorsing a new brand of perfume, and a new brand of men's cologne was endorsed by Tom Selleck.

**Data Collection Procedure.** The subjects for the confirmatory analysis were selected through a systematic area-sampling technique (Churchill 1987) which has been widely used and accepted in survey research (Lovelock et al. 1976; Survey Research Center 1976). In the current study, all census tracts in a small southeastern city were chosen for sampling. Within each tract a number of blocks (depending on the population) were chosen for sampling, excluding commercial blocks and blocks containing parks, churches, or schools. Each interviewer was given a map of the area to be sampled. For each block the interviewer was instructed to randomly select a house and then to conduct interviews at every other house until the quota for that block was filled. The interviewing procedure involved the personal delivery and collection of self-administered questionnaires. A total of 360 questionnaires (180 for each celebrity) were delivered, 289 collected, and 265 found suitable for analysis. The attrition rate was due either to excessive missing data or to obvious response bias. The final sample included 138 usable questionnaires for Linda Evans and 127 for Tom Selleck.

**Confirmatory Factor Analysis.** It is widely recognized that exploratory factor analysis can be quite useful in the early stages of scale development. However, as more knowledge is acquired about the nature of the scale, new data and more rigorous statistical techniques should be applied to confirm or disprove the results obtained in the exploratory stage. Thus, in the second phase of this research, a confirmatory factor analysis model was specified in order to verify the tri-component structure of the scale. The concept of confirmatory factor analysis is as follows: Given a set of observable response variables (the 15 items for attractiveness, expertise, and trustworthiness), this process attempts to determine a smaller set of underlying latent factors (attractiveness, expertise, and trustworthiness dimensions). Using the procedure suggested by Joreskog (1979), the confirmatory factor-analysis model was defined as follows:

$$x = \Lambda\xi + \delta$$

where:

$x$ is a $(15 \times 1)$ column vector of observed variables (the 15 items for source-credibility scale)

$\Lambda$ is a $(15 \times 3)$ column pattern coefficient matrix of $x$ on $\xi$

$\xi$ is a $(3 \times 1)$ column vector of dimensions (attractiveness, expertise, and trustworthiness) derived from the observed variables ($x$)

$\delta$ is a $(15 \times 1)$ column vector of errors of measurement of $x$

$\Phi$ is a $(3 \times 3)$ symmetric covariance matrix of $\xi$

The confirmatory factor analysis model for the 15 source-credibility items is presented in Figure 1. Table 3 presents the input correlation matrices for the 15 source-credibility items of

the Linda Evans and Tom Selleck data sets.

Using the LISREL methodology (Joreskog and Sorbom 1988) to verify the relationship between observable variables and latent constructs, two confirmatory factor-analysis models (Linda Evans and Tom Selleck) were tested separately. The $\chi^2$ statistic was nonsignificant for each model, indicating an adequate fit of the confirmatory model to the data . ($\chi^2_{SELLECK} = 99.60$, df = 87, $p = .168$; $\chi^2_{EVANS} = 109.71$, df = 87, $p = .051$). Further, the plot of the normalized residuals approximated a straight line, indicating that there were no specification errors or departures from normality in the data. In addition, examination of the Q-plots indicated that the normalized residuals had a slope larger than one as compared to the 45-degree line, which is an additional confirmation of the fit of the data to the specified model (Joreskog and Sorbom 1988). The root mean square residual was .048 and .046 for the Selleck and Evans models, respectively.

The reliability estimates of each item are shown in Table 4. As can be seen, the individual items appear to be reliable. Further, the confirmatory factor model for each celebrity explains about 90 percent of the variation for the three dimensions of source credibility, indicating highly reliable dimensions. Finally, the values of $p_{vc}(\xi)$, which are the

FIGURE 1
Confirmatory Factor Analysis Model for the Three Dimensions of the Source-Credibility Scale

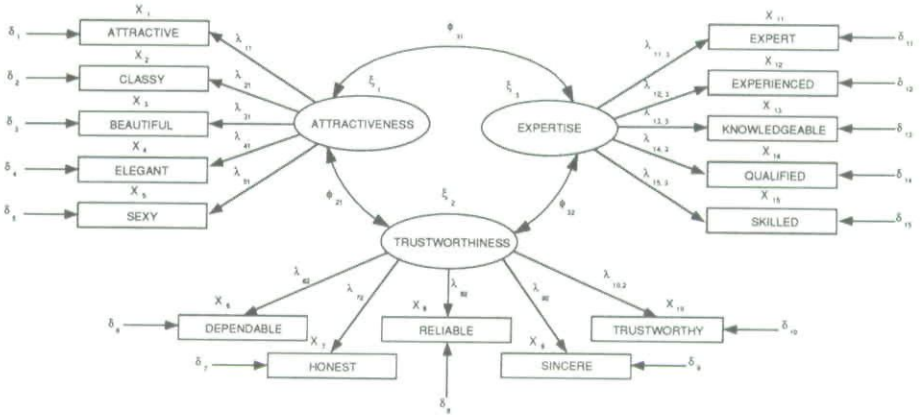



FIGURE 1
Confirmatory Factor Analysis Model for the Three Dimensions of the Source-Credibility Scale

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ | $X_9$ | $X_{10}$ | $X_{11}$ | $X_{12}$ | $X_{13}$ | $X_{14}$ | $X_{15}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $X_1$ | | .595 | .787 | .636 | .715 | .421 | .374 | .411 | .526 | .468 | .326 | .333 | .420 | .377 | .327 |
| $X_2$ | .621 | | .620 | .523 | .498 | .400 | .330 | .402 | .491 | .476 | .247 | .336 | .367 | .332 | .300 |
| $X_3$ | .721 | .678 | | .579 | .706 | .335 | .375 | .366 | .485 | .425 | .354 | .320 | .395 | .354 | .366 |
| $X_4$ | .582 | .666 | .605 | | .530 | .368 | .166 | .312 | .380 | .226 | .290 | .331 | .396 | .293 | .276 |
| $X_5$ | .653 | .561 | .712 | .618 | | .362 | .367 | .426 | .480 | .414 | .449 | .383 | .447 | .423 | .415 |
| $X_6$ | .281 | .357 | .301 | .371 | .288 | | .590 | .694 | .598 | .604 | .333 | .392 | .427 | .354 | .324 |
| $X_7$ | .277 | .295 | .219 | .294 | .243 | .603 | | .589 | .592 | .664 | .295 | .338 | .329 | .360 | .376 |
| $X_8$ | .306 | .314 | .311 | .352 | .304 | .672 | .536 | | .598 | .643 | .286 | .431 | .384 | .314 | .293 |
| $X_9$ | .392 | .395 | .343 | .370 | .371 | .609 | .576 | .647 | | .762 | .375 | .440 | .467 | .461 | .399 |
| $X_{10}$ | .299 | .282 | .236 | .280 | .261 | .632 | .649 | .602 | .758 | | .307 | .403 | .365 | .336 | .375 |
| $X_{11}$ | .188 | .167 | .197 | .147 | .183 | .123 | .058 | .115 | .139 | .167 | | .583 | .660 | .623 | .600 |
| $X_{12}$ | .286 | .278 | .289 | .259 | .292 | .217 | .220 | .317 | .333 | .342 | .636 | | .701 | .563 | .543 |
| $X_{13}$ | .204 | .152 | .232 | .250 | .189 | .191 | .199 | .239 | .201 | .266 | .570 | .585 | | .684 | .640 |
| $X_{14}$ | .165 | .158 | .200 | .228 | .177 | .116 | .109 | .153 | .154 | .129 | .512 | .636 | .715 | | .608 |
| $X_{15}$ | .221 | .270 | .259 | .282 | .281 | .204 | .150 | .283 | .250 | .152 | .488 | .639 | .630 | .650 | |

[1]Above-diagonal entries represent Tom Selleck data. Below-diagonal entries represent Linda Evans data.

average variances extracted by each dimension, are well above the 0.5 cut-off point. Thus, the variance captured by each dimension is significantly higher than the variance due to measurement error, indicating adequate convergent validity for each dimension (Fornell and Larcker 1981). The final tri-component scale is presented in the Appendix.

**Nomological Validity.** Nomological validity investigates the relationship between the scores of a scale and how these scores relate to the measures of other constructs or behaviors. If the suggested relationships between constructs are empirically supported, then it is assumed that the measures of those constructs have a certain degree of nomological validity (Peter 1981).

The nomological validity in this study was tested by relating scores on each dimension of expertise, trustworthiness, and attractiveness to several self-reported behaviors. Specifically, intention to purchase the product, the role of the celebrity in helping the image and the sale of the product, and respondents' liking of and perceived similarity to the source were used as validation behaviors. Given that the validation measures were single-item scales, Pearson Product Moment Correlations were determined to be more appropriate for use than the more rigorous structural equation modeling approach for the assessment of nomological validity. Correlation coefficients were tested for significance at $\alpha = .05$ for the one-tail test. Table 5 shows the correlation coefficients for these analyses.

All three intention-to-purchase items were significantly correlated with the expertise, trustworthiness, and attractiveness dimensions for Tom Selleck. For the Linda Evans sample, inquiring about the brand and considering purchase of the brand were significantly correlated with the three dimensions. The magnitude of the relationship between source trustworthiness and intention to purchase the product for individual consumption was smaller than the other two purchase measures; however, it was still significant at $p < .05$ level.

TABLE 4
Item Reliability, Construct Reliability, Interconstruct Correlation, and Average Variance Extracted for the Three Dimensions of Celebrity Endorser-Credibility Scale

| | *Item Reliability* | |
|---|---|---|
| | *Linda Evans* | *Tom Selleck* |
| **Attractiveness Dimension** | | |
| Attractive | .669 | .799 |
| Classy | .637 | .476 |
| Handsome/Beautiful | .748 | .764 |
| Elegant | .548 | .468 |
| Sexy | .661 | .638 |
| Construct Reliability | .904 | .893 |
| Avg. Var. Extracted | .653 | .629 |
| Construct Correlation Attract/Trust | .477 | .621 |
| **Trustworthiness Dimension** | | |
| Dependable | .674 | .575 |
| Honest | .524 | .558 |
| Reliable | .604 | .596 |
| Sincere | .696 | .704 |
| Trustworthy | .653 | .734 |
| Construct Reliability | .895 | .896 |
| Avg. Var. Extracted | .630 | .633 |
| Construct Correlation Trust/Expert | .319 | .579 |
| **Expertise Dimension** | | |
| Expert | .564 | .590 |
| Experienced | .702 | .587 |
| Knowledgeable | .567 | .767 |
| Qualified | .647 | .616 |
| Skilled | .556 | .557 |
| Construct Reliability | .885 | .892 |
| Avg. Var. Extracted | .607 | .623 |
| Construct Correlation Expert/Attract | .350 | .553 |

Self-report measures of intention to purchase the product for a gift also produced significant correlations across the three source-credibility dimensions. This pattern was true for both the Selleck and the Evans sample. Respondents considered the brands endorsed by Tom Selleck and Linda Evans as distinctive, compared to other brands on the market, and perceived that the celebrity would help the image and the sale of the product. The relationships between these variables and the attractiveness, trustworthiness, and expertise dimensions were significant and in the positive direction.

Finally, as suggested in the literature, likability of the endorser was highly correlated with celebrity endorser's expertise, trustworthiness, and attractiveness. Similar results of a smaller magnitude were also found between respondents' perceived similiarity to the source and the tri-component construct.

**Convergent and Discriminant Validity.** The final step in validating the scale was the determination of its convergent and discriminant validity by way of a multitrait-multimethod matrix (MTMM). For the development of the MTMM, at least three different methods are required to measure each dimension. For this research, Likert and Stapel verions of the source-credibility

TABLE 5
Correlations Among Three Source-Credibility Factors and Several Self-Report Validity Measures[1]

| | *Linda Evans* | | | *Tom Selleck* | | |
|---|---|---|---|---|---|---|
| | *Attractiveness* | *Trustworthiness* | *Expertise* | *Attractiveness* | *Trustworthiness* | *Expertise* |
| **Self-Consumption** | | | | | | |
| Inquire about the brand | .507 | .311 | .474 | .473 | .584 | .520 |
| Consider purchasing the brand | .433 | .210 | .455 | .474 | .568 | .598 |
| Purchase the brand | .374 | .145 | .485 | .467 | .586 | .554 |
| **Gift-Giving** | | | | | | |
| Inquire about the brand | .320 | .197 | .315 | .501 | .451 | .583 |
| Consider purchasing the brand | .290 | .176 | .367 | .489 | .428 | .575 |
| Purchase the brand | .248 | .190 | .404 | .447 | .450 | .561 |
| **Other Validity Measures** | | | | | | |
| Distinctive compared to other brands | .319 | .286 | .364 | .445 | .437 | .526 |
| Celebrity will help the product's sale | .386 | .361 | .492 | .495 | .464 | .584 |
| Celebrity will help the product's image | .301 | .302 | .418 | .427 | .417 | .551 |
| Celebrity is likable | .582 | .598 | .307 | .655 | .661 | .597 |
| Perceived self and source similarity | .257 | .249 | .253 | .364 | .325 | .220 |

[1]All correlation coefficients are significant at $p < .05$ level.

items were developed and used, along with the original semantic differential version to determine if the three dimensions (trait factors)—attractiveness, trustworthiness, and expertise—had convergent and discriminant validity. The celebrity used for this analysis was Linda Evans, promoting a new brand of perfume.

One hundred eight undergraduate students were recruited for this phase of the analysis, with data collected in three stages. In the first stage, each respondent was randomly given one of the three versions (Likert, Stapel, and Semantic Differential) of the scale, along with filler questions. After completing the first version of the questionnaire, the students were given a five-minute break and then given a second version, which also contained filler questions. After a second break, the students completed the third version of the scale. To minimize demand characteristics and response bias, the order of presentation of the items and the versions of the scale were randomly varied among respondents.

The assessment of convergent and discriminant validity of the scale followed the steps outlined by Joreskog (1971) and Bagozzi (1980). The first step for determining the convergent and disciminant validity was to determine if the three traits as measured by the three methods were congeneric. This test assumes that the correlation coefficient among the three traits is equal to unity—that is, the three dimensions of attractiveness, trustworthiness, and expertise are the same. Congenerity is determined by computing the difference between the $\chi^2$ of the following two hypotheses (Joreskog 1971).

$H_1$: $\lambda_1 \ldots \lambda_g$, $\delta_1 \ldots \delta_g$, $p_1$, $p_2$, and $p_3$ are unconstrained.
$H_2$: $p_1 = p_2 = p_3 = 1$

The LISREL model for testing $H_1$ produced a nonsignificant $\chi^2$ ($\chi^2 = 33.07$, d.f. $= 24$, $p < 0.103$). However, the model for $H_2$ produced a highly significant $\chi^2$ value ($\chi^2 = 374.87$, d.f. $= 27$, $p < 0.001$). Based on this evidence, we can reject the assumption of congenerity ($\chi^2 = 341.80$, d.f. $= 3$, $p < 0.001$) and conclude that the attractiveness, trustworthiness, and expertise dimensions do not measure the same trait. Furthermore, the nonsignificant $\chi^2$ for $H_1$ provides evidence for convergent validity. Table 6 provides the MTMM correlations matrix for testing $H_1$ and $H_2$.

Given that the test for congenerity was rejected, the next step was to test for discriminant validity. This involves including a method factor for each trait in $H_1$: The solution for this model yielded a goodness-of-fit of $\chi^2 = 12.06$, d.f. $= 13$ ($p = 0.523$), which represents a good fit of the data to the hypothesized model. Table 7 presents the partitioning of variance due to the trait method and the error factor for each dimension of the source-credibility scale. As can be seen, each of the three methods adequately captured the dimensions of source-credibility. In each case, the variance due to the trait is significantly larger than the variance due to either the method or the error factor. The above analyses provide evidence that the source-credibility scale developed in this study has acceptable convergent and discriminant validity.

## Discussion

Since the original contributions of Hovland and his colleages (Hovland, Janis, and Kelly 1953), the concept of source credibility has been extensively studied in psychology (Berscheid 1966; Chaiken 1979; Johnson, Torcivia, and Popprick 1968; McGinnies and Ward

1980; Mills and Harvey 1972; Ross 1973; Wu and Shaffer 1987), in communication (Applbaum and Anatol 1972; Berlo, Lemert and Mertz 1969; McCroskey 1966; Miller and Baseheart 1969; Whitehead 1968), in marketing, and in advertising (Baker and Churchill 1977; Caballero and Solomon 1984; DeSarbo and Harshman 1985; Kahle and Homer 1985; Mowen and Brown 1981; Wynn 1987). Despite its widespread use, source credibility has not been properly operationalized by means of a reliable and valid scale. In addition, the experimental studies that have used various dimensions of source credibility have not been consistent in their manipulation checks of the experimental variables. The current research has defined the domain of the source-credibility construct and has developed a reliable and valid scale for its measurement.

From a theoretical perspective, the present scale should replace the plethora of single-item measures of attractiveness, expertise, and trustworthiness. By identifying and measuring this tri-component construct, researchers can validly assess the impact of each component of a celebrity endorser's persuasiveness. The consistent use of the same instrument can illuminate the comparison of findings across several studies and can contribute to the source-credibility literature.

The present scale can be adapted to a variety of situations. Researchers in political science can use the scale to investigate the credibility of political candidates. In political campaigns, a candidate's success depends on his/her ability to acquire the voter's trust, approval, and confidence in his/her knowledge and ability. Periodic checks could be performed at various stages in the campaign to evaluate the level of the candidate's credibility. In instructional settings, the scale can be used to evaluate the influence of the instructor's characteristics on student evaluations of the teacher. Finally, in experimental studies of source credibility, the scale can be used to assess the effectiveness of the experimental manipulation. For example, if expertise were the manipulated factor, a post-test administration of the scale should produce a statistically significant difference for low- and high-treatment groups.

Given the large sums of money spent on celebrity advertising, advertisers should use the scale as an integral part of their effectivness testing and tracking. The scale is simple to use and convenient for large-sample administrations. For segmentation strategies, the wisdom of using a celebrity and the spokesperson's effectiveness for different demographic and psychographic consumer groups can be evaluated. This concept also has applications beyond the use of celebrity spokespersons: the scale can be applied effectively in choosing the most appropriate "average consumer" as a spokesperson. Finally, the dimensions of a celebrity endorser's credibility, along with consumer demographics and psychographics, can be used as potential predictors of attitude toward and intention to purchase a specific product.

TABLE 6
Input Correlations for the Multitrait-Multimethod Matrix[1] (n = 108)

| | *Semantic Differential Scale* | | | *Likert Scale* | | | *Stapel Scale* | | |
|---|---|---|---|---|---|---|---|---|---|
| | *Attract* | *Trust* | *Expert* | *Attract* | *Trust* | *Expert* | *Attract* | *Trust* | *Expert* |
| **Semantic Differential Scale** | | | | | | | | | |
| Attract | | | | | | | | | |
| Trust | .397 | | | | | | | | |
| Expert | .402 | .454 | | | | | | | |
| **Likert Scale** | | | | | | | | | |
| Attract | .787 | .410 | .431 | | | | | | |
| Trust | .405 | .717 | .393 | .428 | | | | | |
| Expert | .383 | .419 | .755 | .339 | .357 | | | | |
| **Stapel Scale** | | | | | | | | | |
| Attract | .832 | .440 | .413 | .807 | .393 | .461 | | | |
| Trust | .399 | .856 | .501 | .472 | .783 | .442 | .485 | | |
| Expert | .404 | .437 | .793 | .461 | .443 | .779 | .456 | .477 | |

[1]Attract = Attractiveness; Trust = Trustworthiness; Expert = Expertise

TABLE 7
Partitioning of Variance Due to Trait, Method, and Error

| | *Trait* | *Percent Due to Method* | *Error* |
|---|---|---|---|
| **Semantic Differential Scale** | | | |
| Attractiveness | 0.756 | 0.037 | 0.204 |
| Trustworthiness | 0.790 | 0.027 | 0.183 |
| Expertise | 0.778 | 0.005 | 0.217 |
| **Likert Scale** | | | |
| Attractiveness | 0.912 | 0.072 | 0.016 |
| Trustworthiness | 0.649 | 0.001 | 0.341 |
| Expertise | 0.733 | 0.071 | 0.197 |
| **Stapel Scale** | | | |
| Attractiveness | 0.876 | 0.134 | 0.000[1] |
| Trustworthiness | 0.943 | 0.008 | 0.050 |
| Expertise | 0.828 | 0.003 | 0.169 |

[1]Denotes a parameter fixed at zero to eliminate the Haywood Effect

## Limitations and Research Extensions

The present study has a number of limitations; the recognition of these should help refine future research efforts. With regard to the three dimensions of the scale, the selection of expertise, trustworthiness, and attractiveness was motivated by previous theoretical work and empirical observations, especially in selecting the attractiveness construct. Therefore, one should be cautioned that the quantitative analysis establishes the reliability and validity of the scale rather than discovers their existence. As research findings continue in this area, the existing scale can be expanded or modified.

**Given the large sums of money spent on celebrity advertising, advertisers should use the scale as an integral part of their effectiveness testing and tracking. The scale is . . . convenient for large-sample administrations.**

With regard to generalizability, the findings of this study are limited to the celebrities and product endorsements tested in this research. An interesting retrospection involves the use of celebrities endorsing products that are not considered appropriate for their image. This approach would provide additional credence to the reliability and validity of the scale.

Future research should examine the impact of consumer involvement and confidence in the product as mediating variables in source-credibility research. For example, how does source credibility influence purchase intentions with high-involvement products, as opposed to low-involvement products? Should celebrities and other credible sources be used with high-involvement or low-involvement products? And how does the level of consumer confidence and knowledge about the product mediate the impact of source credibility on intentions to purchase the product?

Additionally, given that the constructs of expertise, trustworthiness, and attractiveness are correlated, an interesting line of research would involve the study and determination of the causal order among these constructs. For example, does a celebrity endorser's attractiveness affect his or her perceived expertise, which in turn, affects the level of trustworthiness? What conditions moderate the order and impact of these variables? Answers to these and other questions should help advertising practitioners identify the most appropriate sources for their clients' products or services. Further, use of this scale can improve the understanding of how consumers in different situations react to different sources and how source credibility influences purchase intentions.

## APPENDIX
## Source-Credibility Scale

**Attractiveness**

Attractive—Unattractive
Classy—Not Classy
Beautiful—Ugly
Elegant—Plain
Sexy—Not sexy

**Trustworthiness**

Dependable—Undependable
Honest—Dishonest
Reliable—Unreliable
Sincere—Insincere
Trustworthy—Untrustworthy

**Expertise**

Expert—Not an expert
Experienced—Inexperienced
Knowledgeable—Unknowledgeable
Qualified—Unqualified
Skilled—Unskilled

## References


Allport, G. and H. Odbert (1936), "Trait-Names: A Psycho-Lexical Study," *Psychological Monographs*, 211.

Anderson, N.H. (1968), "Likableness Ratings of 555 Personality-Trait Words," *Journal of Personality and Social Psychology*, 9 (2), 272-279.

Anderson, K. and T. Clevenger (1963), "A Summary of Experimental Research in Ethos," *Speech Monographs*, 30 (June), 59-78.

Applbaum, Ronald F. and Karl W.E. Anatol (1972), "The Factor Structure of Source Credibility as a Function of the Speaking Situation," *Speech Monographs*, 39 (August), 216-222.

Aronson, Elliot and Philip Worchel (1966), "Similarity versus Liking as Determinants of Interpersonal Attractiveness," *Psychonomic Science*, 5 (4), 157-158.

Bagozzi, Richard P. (1980), *Causal Models in Marketing*, New York: John Wiley and Sons, Inc.

Baker, Michael J. and Gilbert A. Churchill, Jr. (1977), "The Impact of Physically Attractive Models on Advertising Evaluations," *Journal of Marketing Research*, 14 (November), 538-555.

Belk, Russell W. (1982), "Effects of Gift Giving Involvement on Gift Selection Strategies," in *Advances in Consumer Research, Vol. 9*, Andrew Mitchell, ed., Ann Arbor, MI: Association for Consumer Research, 408-411.

Berlo, David K., James B. Lemert, and Robert J. Mertz (1969), "Dimensions for Evaluating the Acceptability of Message Sources," *Public Opinion Quarterly*, 33 (Winter), 563-576.

Berscheid, Ellen (1966), "Opinion Change and Communicator-Communicatee Similarity and Dissimilarity," *Journal of Personality and Social Psychology*, 4 (6), 670-680.

Bowers, John W. and William A. Phillips (1967), "A Note on the Generality of Source Credibility Scales," *Speech Monographs*, 34 (August), 185-186.

*BusinessWeek* (1987), "Nothing Sells Like Sports," (August 31), 48-52.

Caballero, Marjorie, James R. Lumpkin, and Charles S. Madden (1989), "Using Physical Attractiveness as an Advertising Tool: An Empirical Test of the Attraction Phenomenon," *Journal of Advertising Research*, 29 (August/September), 16-22.

Caballero, Marjorie and Paul Solomon (1984), "Effects of Model Attractiveness on Sales Response," *Journal of Advertising*, 13 (1), 17-23, 33.

Chaiken, Shelly (1979), "Communicator Physical Attractiveness and Persuasion," *Journal of Personality and Social Psychology*, 37 (2), 1387-1397.

Churchill, Gilbert A., Jr. (1987), *Marketing Research: Methodological Foundations*, 4th ed., Chicago, IL: The Dryden Press.

Clarke, Keith and Russell W. Belk (1979), "The Effects of Product Involvement and Task Definitions on Anticipated Consumer Effort," in *Advances in Consumer Research, Vol. 6*, William L. Wilkie, ed., Ann Arbor, MI: Association for Consumer Research, 313-318.

Crano, William D. (1970), "Effects of Sex, Response Order, and Expertise in Conformity: A Dispositional Approach," *Sociometry*, 33 (September), 239-252.

Crisci, Richard and Howard Kassinove (1973), "Effects of Perceived Expertise, Strength of Advice, and Environmental Setting on Parental Compliance," *The Journal of Social Psychology*, 89 (2), 245-250.

DeSarbo, Wayne S. and Richard A. Harshman (1985), "Celebrity-Brand Congruence Analysis," *Current Issues and Research in Advertising*, J.H. Leigh and C.R. Martin, Jr., eds., Ann Arbor, MI: Division of Research, Graduate School of Business Administration, The University of Michigan, 17-52.

Fornell, Claes and David F. Larcker (1981), "Evaluating Structural Equation Models with Unobservable Variables and Measurement Error," *Journal of Marketing Research*, 18 (February), 39-50.

Friedman, Hershey and I. Friedman (1976), "Whom Do Students Trust?" *Journal of Communication*, 26 (1), 48-49.

Friedman, Hershey H., Michael J. Santeramo, and Anthony Traina (1979), "Correlates of Trustworthiness for Celebrities," *Journal of Academy of Marketing Science*, 6 (4) 291-299.

Giffin, Kim (1967), "The Contribution of Studies of Source Credibility to a Theory of Interpersonal Trust in the Communication Process," *Psychological Bulletin*, 68 (2), 104-119.

Griffitt, William B. (1966), "Interpersonal Attractions as a Function of Self-Concept and Personality Similarity-Dissimilarity," *Journal of Personality and Social Psychology*, 4 (6), 581-584.

Horai, J.M., N. Naccari , and E. Fatoullah, (1974), "The Effects of Expertise and Physical Attractiveness Upon Opinion Agreement and Liking," *Sociometry*, 37 (4), 601-606.

Hovland, Carl I., Irving K. Janis, and Harold H., Kelley (1953), *Communication and Persuasion*, New Haven, CT: Yale University Press.

Hovland, Carl I. and Walter Weiss (1951), "The Influence of Source Credibility on Communication Effectiveness," *Public Opinion Quarterly*, 15 (Winter), 635-650.

Johnson, Homer H., and Richard Izzett (1972), "The Influence of Source Identification on Attitude Change as a Function of the Type of Communication," *Journal of Social Psychology*, 86 (1), 8-87.

Johnson, Homer H., James M. Torcivia, and Mary Ann Poprick (1968), "Effects of Source Credibility on the Relationship Between Authoritarianism and Attitude Change," *Journal of Personality and Social Psychology*, 9 (2), 179-183.

Joreskog, Karl G. (1971), "Statistical Analysis of Sets of Congeneric Tests," *Psychometrika*, 36 (1), 109-133.

Joreskog, Karl G. (1979), "Basic Ideas of Factor and Component Analysis," in *Advances in Factor Analysis and Structural Equations Models*, K.G. Joreskog and D. Sorbom, eds., Cambridge, MA: Abt Books, 5-20.

Joreskog, Karl G. and Dag Sorbom (1988), *LISREL 7: A Guide to the Program and Applications*, Chicago, IL: SPSS Inc.

Joseph, W. Benoy (1982), "The Credibility of Physically Attractive Communicators: A Review," *Journal of Advertising*, 11 (3), 15-24.

Kahle, Lynn R. and Pamela M. Homer (1985), "Physical Attractiveness of the Celebrity Endorser: A Social Adaptation Perspective," *Journal of Consumer Research*, 11 (March), 954-961.

Kassarjian, Harold H. (1981), "Low Involvement—A Second Look," in *Advances in Consumer Research, Vol. 8*, Kent B. Monroe, ed., Ann Arbor, MI: Association for Consumer Research, 31-34.

Kelman, Herbert C. (1961), "Processes of Opinion Change," *Public Opinion Quarterly*, 33 (Spring), 57-78.

Kelman, H.C. and C.I. Hovland (1953), "Reinstatement of the Communicator in Delayed Measurement of Opinion Change," *Journal of Abnormal and Social Psychology*, 48 (July), 327-335.

Lovelock, Christopher H., Ronald Stiff, David Cullwick, and Ira M. Kaufman (1976), "An Evaluation of the Effectiveness of Drop-Off Questionnaire Delivery," *Journal of Marketing*, 13 (November), 358-364.

Maddux, James E. and Ronald W. Rogers (1980), "Effects of Source Expertness, Physical Attractiveness and Supporting Arguments on Persuasion: A Case of Brains Over Beauty," *Journal of Personality and Social Psychology*, 39 (2), 235-244.

McCroskey, James C. (1966), "Scales for the Measurement of Ethos," *Speech Monographs*, 33, 65-72.

McGinnies, Elliott and Charles D. Ward (1980), "Better Liked Than Right: Trustworthiness and Expertise as Factors in Credibility," *Personality and Social Psychology Bulletin*, 6 (3), 467-472.

McGuire, William J. (1985), "Attitudes and Attitude Change," in *Handbook of Social Psychology, Vol. 2*, Gardner Lindzey and Elliot Aronson, eds., New York: Random House, 233-346.

Miller, Cyndee (1989), "Celebs' Sweet Smell of Success Generates Dollars and Scents," *Marketing News*, (September 25), 8.

Miller, Gerald P. and John Basehart (1969), "Source Trustworthiness, Opinionated Statements, and Response to Persuasive Communication," *Speech Monographs*, 36 (1), (March), 1-7.

Mills, J. and E. Aronson (1965), "Opinion Change as a Function of Communicator's Attractiveness and Desire to Influence," *Journal of Personality and Social Psychology*, 1 (2), 173-177.

Mills, Judson and John Harvey (1972), "Opinion Change as a Function of When Information About the Communicator is Received and Whether He is Attractive or Expert," *Journal of Personality and Social Psychology*, 21 (1), 52-55.

Morrison, Ann M. (1980), "The Boss as Pitchman," *Fortune*, (August 25), 66-73.

Mowen, John C. and Stephen W. Brown (1981), "On Explaining and Predicting the Effectiveness of Celebrity Endorsers," *Advances in Consumer Research: Vol. 8*, Kent M. Moore, ed., Ann Arbor, MI: Association for Consumer Research, 437-441.

Osgood, C.E., G.A. Suci, and P.H. Tannenbaum (1957), *The Measurement of Meaning*, Urbana, IL: University of Illinois Press.

Patzer, Gordon L. (1983), "Source Credibility as a Function of Communicator Physical Attractiveness," *Journal of Business Research*, 11 (2), 229-241.

Peter, Paul J. (1979), "Measurement Reliability: A Review of Psychometric Basics and Recent Marketing Practices," *Journal of Marketing Research*, 16 (February), 18-25.

Peter, Paul J. (1981), "Construct Validity: A Review of Basic Issues and Marketing Practices," *Journal of Marketing Research*, 18 (May), 133-145.

Ross, Joel A. (1973), "Influence of Expert and Peer Upon Negro Mothers of Low Socioeconomic Status," *The Journal of Social Psychology*, 89, 79-84.

Rubin, Vicki, Carol Mager, and Hershey H. Friedman (1982), "Company President versus Spokesperson in Television Commercials," *Journal of Advertising Research*, 22 (August/September), 31-33.

Simon, Herbert W., Nancy N. Berkowitz, and R. John Moyer (1970), "Similarity, Credibility, and Attitude Change: A Review and a Theory," *Psychological Bulletin*, 73 (1), (January), 1-16.

Simpson, Edwin K. and Ruel C. Kahler (1980-81), "A Scale for Source Credibility, Validated in the Selling Context," *The Journal of Personal Selling and Sales Management*, 12 (Fall/Winter), 17-25.

Slinker, Barry H. (1984), "Would You Buy a Burger from This Man? A Car? Some Stocks?" *Madison Avenue*, (April), 52-58.

Steadman, M. (1969), "How Sexy Illustrations Affect Brand Recall," *Journal of Advertising Research*, 9 (February), 15-19.

Sternthal, Brian, Ruby Dholakia, and Clark Leavitt (1978), "The Persuasive Effect of Source Credibility: Tests of Cognitive Response," *Journal of Consumer Research*, 4 (March), 252-260.

Survey Research Center (1976), *Interviewer's Manual, Revised Edition*, Ann Arbor, MI:

The University of Michigan.

Walley, Wayne (1987), "Actors Set Contract Talks," *Advertising Age*, (Dec. 21), 4.

Whitehead, Jack L. (1968), "Factors of Source Credibility," *Quarterly Journal of Speech*, 54 (1), 59-63.

Whittaker, James O. and Robert D. Meade (1968), "Retention of Opinion Change as a Function of Differential Source Credibility: A Cross-Cultural Study," *International Journal of Psychology*, 3 (2), 103-108.

Widgery, Robin N. and Richard S. Ruch (1981), "Beauty and the Machiavellian," *Communication Quarterly*, 29 (Fall), 297-301.

Wilding, John and Raymond A. Bauer (1968), "Consumer Goals and Reactions to a Communication Source," *Journal of Marketing Research*, 5 (February), 73-77.

Woodside, Arch G. and J. William Davenport, Jr. (1974), "The Effect of Salesman Similarity and Expertise on Consumer Purchasing Behavior," *Journal of Marketing Research*, 11 (May), 198-202.

Woodside, Arch G. and J. William Davenport, Jr. (1976), "Effects of Price and Salesman Expertise on Customer Purchasing Behavior," *Journal of Business*, 49 (January), 51-59.

Wynn, George W. (1987), "The Effects of a Salespersons' Credibility on Other Salespersons and Sales Managers," *Developments in Marketing Science, Vol. 10*, Jon M. Hawes and George B. Glisan, eds., Bal Harbour, FL: Academy of Marketing Science, 353-358.

Wu, Chenghuan and David R. Shaffer (1987), "Susceptibility to Persuasive Appeals as a Function of Source Credibility and Prior Experience with the Attitude Object," *Journal of Personality and Social Psychology*, 52 (4), 677-688.

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